EE 4940 Fall 2026

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Cumulative Reflection Assignment

My experience at Iowa State University was not what I expected and has vastly changed how I think about engineering. When I started college, I always thought of engineers as being problem solvers using complex equations. As I've taken my engineering courses and done my engineering projects, I realized engineering is a lot more than just finding what "x" is equal to. Engineers need to be able to understand problems, communicate with people, be able to do research, and be able to adapt when problems arise. My experiences as an engineering student with courses and projects have exposed me to a broader lens of what it means to be an engineer.

One of the biggest areas where I've needed to grow was learning how to solve problems without a solution that is clear. Projects such as my signal generator required me to take things I've learned from different courses and apply them to solving the problem. I needed to be able to do circuit analysis while also learning about how different types of circuits affect each other when being amplified. When the expected results did not occur, I had to go back and find out where the problems were. I ended up needing to change my resistor values and how they were connected in my circuit to get the right outputs. Another big project was on the Cybot, which took a lot of time and communication with my partners to solve every problem we ran into. Communication is extremely important as people can approach problems differently, which can lead to different outcomes. Doing these projects has made me realize that engineers have a big responsibility when designing solutions. People need solutions to work, but as an engineer, you need to take into account how these solutions affect others.

Learning engineering does not stop at the classroom doors. Learning and truly understanding the material takes a lot of time outside lectures. Resources such as textbooks, examples, and working with classmates help fill in the gaps of knowledge. My work with Matlab is a great example, as it took working with classmates to properly understand how the syntax works. Working with more simulations and learning about expected results helped me understand the relationship between real and simulated results. Sometimes, being an engineer does not mean just knowing things. It means that you know where to go to find reliable information when I need to know something.

Outside of classes, the use of career fairs has helped me better understand the different opportunities available to me as an engineering student and what employers are looking for. Talking to recruiters has helped me understand what skills and experiences are valuable outside of what I learn in my classes. It has shown me what I need to work on before starting my engineering career. Experiences like this have helped me see how taking advantage of opportunities outside of school is important and how I can learn from them.

If I could restart my undergraduate experience, there would be many things I would change, but one of the most important would be prepping for exams earlier. I constantly underestimated how long it would take to understand a topic and would pay the price for it on the exam. I have learned that waiting makes learning difficult topics a lot harder as there isn't much time left to try and experiment to learn why things are the way they are. I would also try and reach out more to people for help. It's nice to work alone, but there are times when someone else's perspective helps more.

A recent example of learning new material has been my work with microwave engineering. Concepts such as Smith charts, impedance matching, and circuit simulation made me tackle problems differently. Instead of just solving equations, I had to understand how the equations represented by physical behavior of circuit components worked and how they appeared in simulations. I had to work through examples, practice using the smith chart, and compare my solutions with ADS. This process was sometimes very frustrating as the expected results were not always produced. However, troubleshooting has helped me understand the material better than just trying to memorize the equations.

My learning strategies have also changed throughout college. Earlier in my education, I always relied on memorizing equations and processes instead of trying to understand why things are the way they are. As my courses became more advanced, I could not do that as much. Now, I try and understand what everything in an equation means and do things step by step. I sometimes do things like comparing calculations to simulations, which may take more effort, but it helps me better understand the material and apply it.

As I work towards being an engineer, I know that there are so many things I need to improve. I want to continue working on my technical skills while also becoming better at communication, teamwork, and problem solving. I also want to become better at working with technology I do not understand without it needing to be explained. My goal is to solve problems with the thought that I may not be able to immediately solve the problem, but I can figure it out. Continuing to learn through the rest of my educational experience and professional experiences will always be important to my development as an engineer.

Overall, my experience at Iowa State has taught me that becoming an engineer is not just about taking courses. It's about learning how to solve problems, learn better, adapt, and communicate with others. My coursework and project experiences have given me a great technical foundation, while working outside the classroom has helped me apply those skills in other situations. As I work towards my engineering career, I expect that learning and adapting will be one of the most important skills I take with me from Iowa State University.